# A much-larger ideal quotient not isomorphic to $\ell_\infty/c_0$

Lane 19 packet for arXiv:1909.10417

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## Source and target

The source is Miek Messerschmidt, *A family of quotient maps of $\ell^\infty$ that do not admit uniformly continuous right inverses*, arXiv:1909.10417 [2]. For ideals $\mathcal{I}, \mathcal{J}$ on $\mathbb{N}$, the source writes $\mathcal{I} \Subset \mathcal{J}$ when $\mathcal{I} \subsetneq \mathcal{J}$ and there are nonempty pairwise disjoint sets $(D_j) \subseteq \mathcal{I}$ such that $\bigcup_{j \in K} D_j \in \mathcal{J} \setminus \mathcal{I}$ for every infinite $K \subseteq \mathbb{N}$. It proves that if $\mathcal{I} \Subset \mathcal{J}$, then the quotient map

$$\mathcal{J}c_0 \longrightarrow \mathcal{J}c_0/\mathcal{I}c_0$$

has no uniformly continuous right inverse.

After proving abundance of much-larger inclusions, the paper remarks that one could conjecture that $\mathcal{J}c_0/\mathcal{I}c_0$ is always isomorphic to $\ell_\infty/c_0$.

number of such subspaces and thereby we provide a large family of examples of quotient maps of $\ell^\infty$ that do not admit uniformly continuous (or Lipschitz) right inverses, on top of the previously known two such examples, Examples 1.4 and 1.5, above.

In Section 2 we provide the definition of an ideal $\mathbf{i}$ in $\mathbb{N}$, the meaning of $\mathbf{i}$-convergence, and we introduce the notation $\mathbf{i}c_0$ by which we will denote the closed subspace of $\ell^\infty$ of all elements that $\mathbf{i}$-converge to zero.

In Section 3 we define a strict partial order that we call 'much larger inclusion' (cf. Definition 3.1) for general ideals $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{N}$. We use the notation $\mathbf{i} \Subset \mathbf{j}$. Through application of Kalton's Monotone Transfinite Sequence Theorem (Theorem 1.7), we arrive at our main tool, Corollary 3.8, stating that if $\mathbf{i} \Subset \mathbf{j}$, then the quotient map $\mathbf{j}c_0 \to \mathbf{j}c_0/\mathbf{i}c_0$ admits no uniformly continuous right inverse.

We also include some general remarks in Section 3 on the much larger inclusion relation. Firstly, much larger inclusion is distinct from set inclusion (cf. Remark 3.2). Further, for any ideals $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{N}$ with $\mathbf{i} \Subset \mathbf{j}$, there necessarily exists uncountably many incomparable ideals $\mathbf{k}$ in $\mathbb{N}$, satisfying $\mathbf{i} \Subset \mathbf{k} \Subset \mathbf{j}$ (cf. Proposition 3.3). Also, there exist maximal $\Subset$-chains (cf. Definition 3.4) of ideals in $\mathbb{N}$ with the ideal $\mathbf{f}$ of all finite subsets of $\mathbb{N}$ as smallest element, and $2^{\mathbb{N}}$ as largest element and such maximal chains are necessarily uncountable as well (cf. Proposition 3.5). Therefore, not unexpectedly, much larger inclusions occur in some abundance within the set of all ideals in $\mathbb{N}$. Since there appears to be nothing special about any particular instance of a much larger inclusion, for any ideals $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{N}$ with $\mathbf{i} \Subset \mathbf{j}$, we could conjecture that the quotient space $\mathbf{j}c_0/\mathbf{i}c_0$ might always be isomorphic to $\ell^\infty/c_0$.

In Section 4 we introduce some classical ideals in $\mathbb{N}$: The ideal $\mathbf{f}$ of all finite subsets of $\mathbb{N}$, the ideal $\mathbf{r}$ of all subsets of $\mathbb{N}$ whose reciprocal series converge, and the ideals $\mathbf{d}, \mathbf{b}, \mathbf{buck}$ and $\mathbf{t}$ of all subsets of $\mathbb{N}$ whose respective, densities, Banach densities, Buck densities, and, tile densities are zero. We show that Buck density and tile density are actually identical, and, since tile density is easier to handle, we subsequently only consider tile density.

The final section of the paper is somewhat technical, and is devoted to establishing the following much larger inclusions of the mentioned classical ideals in $\mathbb{N}$.

This packet gives a counterexample to that conjectural remark.

## Definitions

For an ideal $\mathcal{J}$ on $\mathbb{N}$,

$$\mathcal{J}c_0 = \{x \in \ell_\infty : \{n \in \mathbb{N} : |x_n| \geq \varepsilon\} \in \mathcal{J} \text{ for every } \varepsilon > 0\}.$$

For $\mathcal{J} = \mathrm{Fin}$ this is the usual $c_0$.

A Banach space is Grothendieck if every weak-star convergent sequence in the dual is weakly convergent. We use the classical facts that $\ell_\infty$ is Grothendieck, quotients and complemented subspaces of Grothendieck spaces are Grothendieck, and $c_0$ is not Grothendieck.

## Proof intuition

The source's much-larger relation only requires one countable family of small pieces whose infinite unions become large. We put that witness inside a single infinite block $D$. Separately, we add infinitely many other infinite blocks $A_m$ to the larger ideal, but only finitely many at a time. Those $A_m$ blocks survive in the quotient as a block coordinate system: ultrafilter limits on the blocks define a bounded projection from $\mathcal{J}c_0/c_0$ onto a copy of $c_0$. Since $\ell_\infty/c_0$ is Grothendieck and hence cannot contain complemented $c_0$, the quotient cannot be isomorphic to $\ell_\infty/c_0$.

**Theorem 1.** *There is an ideal $\mathcal{J}$ on $\mathbb{N}$ such that* $\mathrm{Fin} \Subset \mathcal{J}$*, but $\mathcal{J}c_0/c_0$ is not isomorphic to $\ell_\infty/c_0$.*

*Proof.* Partition $\mathbb{N}$ into pairwise disjoint infinite sets

$$\mathbb{N} = D \sqcup A_1 \sqcup A_2 \sqcup \cdots .$$

Define

$$\mathcal{J} = \{S \subseteq \mathbb{N} : \text{ for some } n,\ S \setminus D \subseteq A_1 \cup \cdots \cup A_n\}.$$

This is a proper ideal: it is hereditary, closed under finite unions, contains all finite sets, and does not contain $\bigcup_{m\geq 1} A_m$.

We first verify $\mathrm{Fin} \Subset \mathcal{J}$. Choose pairwise disjoint singletons $D_j = \{d_j\}$ inside $D$. Each $D_j$ belongs to Fin. If $K \subseteq \mathbb{N}$ is infinite, then $\bigcup_{j\in K} D_j$ is an infinite subset of $D$, hence belongs to $\mathcal{J} \setminus \mathrm{Fin}$. Thus $\mathrm{Fin} \Subset \mathcal{J}$.

It remains to show that $\mathcal{J}c_0/c_0$ cannot be isomorphic to $\ell_\infty/c_0$. For each $m$, fix a free ultrafilter on $A_m$ and let

$$\varphi_m : \ell_\infty(A_m) \to \mathbb{R}$$

be the corresponding ultrafilter-limit functional. Then $\|\varphi_m\| = 1$, $\varphi_m(1_{A_m}) = 1$, and $\varphi_m$ vanishes on $c_0(A_m)$.

Define a bounded linear operator $R : \mathcal{J}c_0 \to \ell_\infty$ by

$$(Rx)(n) = \begin{cases} \varphi_m(x|_{A_m}), & n \in A_m, \\ 0, & n \in D. \end{cases}$$

We claim that $R(\mathcal{J}c_0) \subseteq \mathcal{J}c_0$. Let $x \in \mathcal{J}c_0$ and write $b_m = \varphi_m(x|_{A_m})$. If $|b_m| \geq \varepsilon$, then $\sup_{n\in A_m} |x_n| \geq \varepsilon$. Hence

$$\{m : |b_m| \geq \varepsilon\}$$

is finite, because the set $\{n : |x_n| \geq \varepsilon\}$ belongs to $\mathcal{J}$ and therefore meets only finitely many of the blocks $A_m$ outside $D$. Thus $(b_m) \in c_0$, so $Rx \in \mathcal{J}c_0$.

Moreover, $R$ annihilates $c_0$: if $x \in c_0$, then $x|_{A_m} \in c_0(A_m)$ for every $m$, and therefore $\varphi_m(x|_{A_m}) = 0$. Thus $R$ descends to a bounded operator

$$\overline{R} : \mathcal{J}c_0/c_0 \longrightarrow \mathcal{J}c_0/c_0.$$

It is a projection. Indeed, if $x$ is already constant with value $b_m$ on each block $A_m$ and zero on $D$, then $\varphi_m(x|_{A_m}) = b_m$.

The range of $\overline{R}$ is isometric to $c_0$. For $a = (a_m) \in c_0$, let $x_a \in \mathcal{J}c_0$ be zero on $D$ and equal to $a_m$ on $A_m$. The map

$$T : c_0 \to \mathcal{J}c_0/c_0, \qquad T(a) = [x_a]$$

has range equal to $\operatorname{ran} \overline{R}$. Its quotient norm is exactly $\|a\|_\infty$: the upper bound is clear by taking the zero element of $c_0$, while for any $z \in c_0$ and any $m$, the set $A_m$ is infinite, so points in $A_m$ can be chosen with $|z_n|$ arbitrarily small; hence

$$\|x_a - z\|_\infty \geq |a_m| - \delta$$

for every $m$ and every $\delta > 0$. Taking the supremum over $m$ and then $\delta \downarrow 0$ gives $\operatorname{dist}(x_a, c_0) \geq \|a\|_\infty$. Therefore $T$ is an isometry onto the complemented range of $\overline{R}$.

Consequently $\mathcal{J}c_0/c_0$ contains a complemented subspace isomorphic to $c_0$.

On the other hand, $\ell_\infty/c_0$ is Grothendieck. Indeed, $\ell_\infty$ is Grothendieck by Grothendieck's theorem, and quotients of Grothendieck spaces are Grothendieck. Complemented subspaces of Grothendieck spaces are also Grothendieck. But $c_0$ is not Grothendieck: the unit vector basis of $\ell_1 = (c_0)^*$ is weak-star null on $c_0$, but it is not weakly null, since the constant-one functional in $\ell_\infty = (\ell_1)^*$ takes value 1 on every basis vector.

Thus $\ell_\infty/c_0$ cannot contain a complemented subspace isomorphic to $c_0$. Since $\mathcal{J}c_0/c_0$ does contain such a subspace, $\mathcal{J}c_0/c_0$ is not isomorphic to $\ell_\infty/c_0$. □

## Verification and limitations

The construction is entirely in ZFC and uses no special set-theoretic axiom. It refutes the source's conjectural isomorphism remark for arbitrary ideals with $\mathcal{I} \Subset \mathcal{J}$. It does not address whether the specific classical examples in the source, such as the density-zero, Banach-density-zero, tile-density-zero, or reciprocal-series ideals, have quotients isomorphic to $\ell_\infty/c_0$.

Bounded novelty checks were performed on 2026-06-20 against the local run indexes, local parsed arXiv corpus, and web searches for the exact conjectural phrase and close variants. The searches found the source arXiv page and no prior answer to this specific conjectural remark.

## References

## References


[1] A. Grothendieck, *Sur les applications lineaires faiblement compactes d'espaces du type $C(K)$*, Canadian Journal of Mathematics 5 (1953), 129–173.

[2] M. Messerschmidt, *A family of quotient maps of $\ell^\infty$ that do not admit uniformly continuous right inverses*, arXiv:1909.10417, 2019.